

Undergraduate Topics in Mathematics 5
Abstract Algebra

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Abstract Algebra

1.1 Relations and functions

- **Relations:** Let S and T be sets. A **relation** from S to T is a subset ρ of $S \times T$, where

$$S \times T = \{(s, t) : s \in S, t \in T\}.$$

If $s \in S$ and $t \in T$, then we write $s\rho t$ if $(s, t) \in \rho$, otherwise we write $s\not\rho t$. In particular, a relation on S is a relation from S to S .

- **Properties of relations:** Let ρ be a relation on S ,
 - **Reflexive:** ρ is reflexive if $a\rho a$ for all $a \in S$
 - **Symmetric:** ρ is symmetric if $a\rho b$ implies $b\rho a$
 - **Transitive:** ρ is transitive if $a\rho b$ and $b\rho c$ implies that $a\rho c$

Note: These properties are defined only for relations from a set to itself.

- **Equivalence relation:** An **equivalence relation** on a set S is a relation that is reflexive, symmetric, and transitive. We use $\sim$ to denote an equivalence relation.

As an example, on $\mathbb{R}$, let us say that $a \sim b$ if and only if $a + b$ is even. First, consider $a, b \in \mathbb{R}$. Observe that

$$a \sim a$$

since whether or not a is even or odd, the sum is always even. So, $\sim$ is reflexive. Next, we notice that if we assume $a \sim b$ (so $a + b$ is even), it immediately follows that $b + a$ is even. So, $b \sim a$ and $\sim$ is symmetric.

Now, let $a, b, c \in \mathbb{R}$. Assume that

$$a + b, b + c$$

are both even. Hence,

$$a + b + b + c = a + 2b + c = a + c + 2b$$

is also even. Notice that $2b$ is even, so we consider two cases, either $a + c$ is odd and $\sim$ is not transitive or $a + c$ is even. If $a + c$ is odd, then we have $a + c = 2\ell + 1$, and

$$a + c + 2b = 2\ell + 1 + 2b = 2(\ell + b) + 1,$$

which is odd. But, we said that $a + c + 2b$ is even, a contradiction. Thus, $a \sim c$ and $\sim$ is transitive.

Since $\sim$ is reflexive, symmetric, and transitive, it is an equivalence relation. ■

- **Equivalence class:** Let $\sim$ be an equivalence relation on a set S . If $a \in S$, then the **equivalence class** of a , denoted $[a]$, is the set $\{b \in S : a \sim b\}$
- **Partition:** Let S be a set, and let T be a set of nonempty subsets of S . We say that T is a **partition** of S if every $a \in S$ lies in exactly one set in T .
- **Theorem 1.1:** Let S be a set and $\sim$ be an equivalence relation on S . Then the equivalence classes with respect to $\sim$ form a partition of S . In particular, if $a \in S$, then $a \in [a]$, and $a \in [b]$ if and only if $[a] = [b]$.

Proof. Assume S is a set, and $\sim$ is an equivalence relation on S . To show that the equivalence relations with respect to $\sim$ form a partition of S , we must show that the equivalence relations are a set of nonempty pairwise disjoint subsets of S .

Let $a \in S$. By reflexivity of $\sim$, $a \sim a$. Thus, $a \in [a]$ and the equivalence classes are nonempty. Now, assume that

$$d \in [a] \cap [b].$$

Thus,

$$a \sim d, \quad b \sim d.$$

Now, if $c \in [a]$, then $a \sim c$. Notice that by transitivity,

$$d \sim a, a \sim c \implies c \sim d.$$

Similarly,

$$c \sim d, d \sim b \implies c \sim b.$$

Thus, $c \in [b]$. Since $c \in [a]$ implies that $c \in [b]$,

$$[a] \subseteq [b].$$

A similar argument, except starting with $c \in [b]$ yields that

$$[b] \subseteq [a].$$

Hence, if $d \in [a] \cap [b]$, it follows that $[a] = [b]$.

Therefore, if $a \in S$, then $a \in [a]$, and if $a \in [b]$, then $[b] = [a]$. Conversely, if $[a] = [b]$, then $a \in [a] = [b]$. ■

- **The partition:** So, now we know that for a equivalence relation $\sim$ on S ,

$$\sim \subseteq S \times S,$$

if $a \in S$, then the equivalence class of a is the set

$$[a] = \{b \in S : a \sim b\}.$$

Further, $b \in [b]$, and so it must be that $[b] = [a]$. In other words, the representative element is not unique for each class.

- **Functions:** Let S and T be any sets. A **function**

$$\alpha : S \rightarrow T$$

is a rule assigning to each $s \in S$ an element $\alpha(s) \in T$.

- **Requirements of a function:** For a relation $f \subseteq A \times B$ to be called a **function** from A to B , it must satisfy

1. **Existence / totality:** For all $a \in A$, there exists a $b \in B$ such that $(a, b) \in f$

$$\forall a \in A, \exists b \in B : (a, b) \in f.$$

2. **Uniqueness / well-definedness:** Each input in A has exactly one output in B

$$\forall a \in A, \forall b_1, b_2 \in B, ((a, b_1) \in f \wedge (a, b_2) \in f) \implies b_1 = b_2.$$

- **Properties of functions:** Consider a function $\alpha : S \rightarrow T$,
 - **Injective:** α is injective if $\alpha(s_1) = \alpha(s_2)$ implies $s_1 = s_2$ for all $s_1, s_2 \in S$
 - **Surjective:** α is surjective if for all $t \in T$, there exists an element $s \in S$ such that $\alpha(s) = t$
 - **Bijective:** α is bijective (invertible) if it is both injective and surjective

We call a bijective function a **bijection**, which is therefore also a **injection** and a **surjection**.

Notice that the inverse of a function only satisfies existence / uniqueness if it is both injective (uniqueness) and surjective (existence).

- **Composition:** Let R, S , and T be sets, and let $\alpha : R \rightarrow S$ and $\beta : S \rightarrow T$ be functions. Then, the **composition** $\beta \circ \alpha$, or simply $\beta\alpha$, is the function from R to T given by

$$(\beta\alpha)(r) = \beta(\alpha(r))$$

for all $r \in R$. Note that $\alpha\beta$ and $\beta\alpha$ are likely different functions.

- **Theorem 1.2: Properties of the composition:** Let $\alpha : R \rightarrow S$, $\beta : S \rightarrow T$, and $\gamma : T \rightarrow U$ be functions. Then,
 1. $(\gamma\beta)\alpha = \gamma(\beta\alpha)$
 2. If α and β are injective, then so is $\beta\alpha$
 3. If α and β are surjective, then so is $\beta\alpha$
 4. If α and β are bijective, then so is $\beta\alpha$

Proof. First, notice that

$$\begin{aligned} (\gamma\beta)\alpha &= \gamma(\beta\alpha), \\ \gamma(\beta\alpha) &= \gamma(\beta\alpha). \end{aligned}$$

Hence, the first property holds. Now, consider $r_1, r_2 \in R$, and assume that

$$\beta(\alpha(r_1)) = \beta(\alpha(r_2)).$$

Since β is injective, $\alpha(r_1) = \alpha(r_2)$, and since α is injective, $r_1 = r_2$. Thus, $\beta\alpha$ is injective.

Next, consider $t \in T$. Since β is surjective, there exists a $s \in S$ such that $\beta(s) = t$. But, α is surjective, so there exists an $r \in R$ such that $\alpha(r) = s$. Thus,

$$(\beta\alpha)r = \beta(\alpha(r)) = \beta(s) = t.$$

Since we showed that $\beta\alpha$ is both injective and surjective, it follows that $\beta\alpha$ is bijective. ■

- **Theorem 1.3: Existence of the inverse:** Let $\alpha : S \rightarrow T$ be a bijection. Then, there exists a bijection $\beta : T \rightarrow S$ such that $(\beta\alpha)(s) = s$ for all $s \in S$ and $(\alpha\beta)(t) = t$ for all $t \in T$.

Proof. Since α is bijective, for all $t \in T$, there exist a unique $s \in S$ such that $\alpha(s) = t$. Now, define $\beta : T \rightarrow S$ such that for that unique s that maps to t ,

$$\beta(t) = s.$$

Then,

$$(\beta\alpha)(s) = \beta(\alpha(s)) = \beta(t) = s.$$

Now, for any $t \in T$, choose $s \in S$ such that $\alpha(s) = t$. By that same property of β ,

$$\beta(t) = s,$$

so

$$(\alpha\beta)(t) = \alpha(\beta(t)) = \alpha(s) = t.$$

Now, we just need to show that β is a bijection. First, assume that $\beta(t_1) = \beta(t_2)$. Then,

$$\begin{aligned}\alpha(\beta(t_1)) &= t_1, \\ \alpha(\beta(t_2)) &= t_2.\end{aligned}$$

But, $\beta(t_1) = \beta(t_2)$, so

$$\alpha(\beta(t_2)) = t_2 = \alpha(\beta(t_1)) = t_1.$$

Thus, β is injective. Next, $s \in S$, we have that

$$\beta(\alpha(s)) = s.$$

Thus, there exists a $t \in T$ such that $\beta(t) = s$ ($t = \alpha(s)$), so β is surjective. ■

- **Permutation:** A permutation of a set S is a bijection from S to S .
- **Binary operation:** Let S be a set. Then, a **binary operation** on S is a function from $S \times S$ to S .

1.2 The Integers and Modular Arithmetic

1.2.1 Induction

- **Property 2.1: Well ordering axiom:** If S is a nonempty set of positive integers, then S has a smallest element.
- **Theorem 2.1: Mathematical induction:** Suppose that for each positive integer n , we have a proposition $P(n)$. Further suppose that
 1. $P(1)$ is true, and
 2. for each $n \in \mathbb{N}$, if $P(n)$ is true, then so is $P(n+1)$

Then, $P(n)$ is true for every positive integer n .

Proof. We proceed by contradiction. Recall that to show a statement $P \implies Q$ is true, it is merely sufficient to show that $P \wedge \neg Q$ is impossible. Thus, if we assume $\neg Q$, and yield a contradiction, $P \wedge \neg Q$ cannot occur and thus the statement is true.

So, we assume that $P(n)$ is not true for every positive integer n . Thus, there is a nonempty set $S \subseteq \mathbb{N}$ such that for every element $n \in S$, $P(n)$ is false. By the well-ordering axiom, there exists a smallest element in S , call this element k .

Observe that the statement assumes that $P(1)$ is true, and for each $n \in \mathbb{N}$, if $P(n)$ is true, then so is $P(n+1)$. So, $k > 1$. Further,

$$k-1 \notin S,$$

since if $k-1 \in S$, $k-1+1 = k \notin S$, but k is indeed a member of S . However, since $k-1 \notin S$, $P(k-1)$ is true, and thus $P(k-1+1) = P(k)$ is true, a contradiction.

Therefore, if $P(1)$ is true, and for each $n \in \mathbb{N}$, if $P(n)$ is true, $P(n+1)$ is also true, then $P(n)$ is true for every positive integer n . ■

Let's consider an example. Suppose we claim that for every positive integer n ,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

First, we observe that

$$1^2 = \frac{1(1+1)(2(1)+1)}{6} = 1.$$

So, $P(1)$ is true. Next, assume that $P(n)$ is true, where

$$P(n) = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Now, observe that

$$\begin{aligned} P(n+1) &= 1^2 + 2^2 + \cdots + n^2 + (n+1)^2 = P(n) + (n+1)^2 \\ &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{n(2n+1)}{6} + (n+1) \\ &= \frac{n(2n+1) + 6(n+1)}{6} = \frac{2n^2 + n + 6n + 6}{6} = \frac{2n^2 + 7n + 6}{6}. \end{aligned}$$

From here, we can get that

$$\frac{2n^2 + 7n + 6}{6} = \frac{(n+2)(2n+3)}{6}.$$

So,

$$\frac{P(n)}{n+1} + (n+1) = \frac{(n+2)(2n+3)}{6}.$$

Thus,

$$P(n) + (n+1)^2 = P(n+1) = \frac{(n+1)((n+1)+1)(2(n+1)+1)}{6}$$

Thus, $P(n)$ implies $P(n+1)$, and P therefore holds for every positive integer n .

- **Binomial theorem:** Let a and b be real numbers, and n a positive integer. Then,

$$\begin{aligned} (a+b)^n &= a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{n-1}ab^{n-1} + b^n \\ &= \sum_{k=0}^{n-1} \binom{n}{k} a^{n-k} b^k. \end{aligned}$$

Proof. prove by induction on n .

- **Strong induction:** Suppose that, for each positive integer n , we have a proposition $P(n)$. Further suppose that
 1. $P(1)$ is true, and
 2. for each integer $n > 1$, if $P(k)$ is true for every $k < n$, then $P(n)$ is true.

Then, $P(n)$ is true for every positive integer n .

Proof. Assume that the two conditions hold, but $P(n)$ is not true for every positive integer n . Then, let S be the set of integers in which $P(s)$ is false. Since $S \subseteq \mathbb{N}$ and is nonempty, here is a smallest element j by the well-ordering axiom. Call this smallest element ℓ .

Note that since $P(1)$ is true, $\ell > 1$. Since ℓ is the smallest element in S , and therefore the first element that makes the proposition false, it must be that $P(k)$ is true for all $k < \ell$. But, that would imply that $P(\ell)$ is in fact true, a contradiction.

Thus, there is no such set of elements S that make P false, so $P(n)$ true for all $n \in \mathbb{N}$.

1.2.2 Some Number Theory

- **Theorem 2.4 (Division algorithm):** Let $a, b \in \mathbb{Z}$ with $b > 0$. Then, there exists unique integers q and r such that

$$a = bq + r, \quad 0 \leq r < b.$$

We call q and r the **quotient** and **remainder** respectively.

Proof. First, let

$$S = \{a - bt : t \in \mathbb{Z}, a - bt \geq 0\}.$$

First, if $0 \in S$, then $a - bq = 0$ for some $q \in \mathbb{Z}$, and so

$$a = bq + 0$$

and the theorem holds. So, $S \subseteq \mathbb{N}$. We first claim that S is nonempty. Let

$$t = -|a|.$$

Then,

$$a - bt = a + |a|b.$$

If $a \geq 0$, then $a + |a|b \geq 0$ since $b > 0$. If $a \leq 0$ then

$$a + |a|b = a - ab = a(1 - b).$$

Notice that since $b > 0, b \in \mathbb{Z}$, it must be that $b \geq 1$. Thus,

$$1 - b \leq 0.$$

So, $a(1 - b) \geq 0$. Thus, we have shown the existence of at least one t such that $a - bt \in S$, and so S is nonempty.

Since S is a nonempty subset of the natural numbers, we have a least element by the well-ordering axiom, say

$$r = a - bq.$$

Recall that $r \geq 0$ by definition. Suppose that $r \geq b$, then $0 \leq r - b < r$. But,

$$r - b = a - bq + b = a - b(q + 1) < r,$$

a contradiction, since r is the smallest element. Note that

$$r - b = a - b(q + 1)$$

is indeed a member of S , since $r - b \geq 0$. Thus,

$$a = bq + r, \quad 0 \leq r < b.$$

For uniqueness, suppose that

$$a = bq_1 + r_1 = bq_2 + r_2$$

with $q_i, r_i \in \mathbb{Z}$ and $0 \leq r_i < b$. Then,

$$b(q_1 + q_2) = r_1 - r_2.$$

If we take the absolute value,

$$|b(q_1 + q_2)| = |r_1 - r_2|.$$

But, $b > 0$, so

$$b|q_1 - q_2| = |r_2 - r_1|.$$

Notice that since $q_1, q_2 \in \mathbb{Z}$, $q_1 - q_2 \in \mathbb{Z}$, so if $q_1 \neq q_2$,

$$b|q_1 - q_2| \geq b.$$

But, $0 \leq r_1, r_2 < b$, so

$$|r_2 - r_1| < b,$$

and $|r_2 - r_1| = b|q_1 - q_2|$, which we just said was $\geq b$, a contradiction. Hence, $q_1 = q_2$. In that case,

$$b(q_1 - q_2) = b(0) = r_2 - r_1.$$

Thus, $r_1 = r_2$. ■.

- **Divides:** Let a and b be integers, we say that a divides b if there exists an integer c such that $b = ac$, in which case we write

$$a \mid b.$$

If no such c exists, we write $a \nmid b$

- **Lemma 2.1:** Let $a, b, c \in \mathbb{Z}$. Then,
 1. if $a \mid b$ and $b \mid c$, then $a \mid c$
 2. if $a \mid b$ and $b \neq 0$, then $a \leq |b|$
 3. if $a \mid b$ and $a \mid c$, then $a \mid (bu + cv)$ for any $u, v \in \mathbb{Z}$

Proof. First, assume that $a, b, c \in \mathbb{Z}$. (1). Assume that $a \mid b$, and $b \mid c$. Then,

$$b = a\ell, \quad c = bk.$$

If $k = 0$, then $c = 0$, and $a \mid c$. So, assume that $k \neq 0$. Then,

$$b = \frac{c}{k},$$

and so

$$\frac{c}{k} = a\ell.$$

Hence, $c = a(\ell k)$, and $a \mid c$.

(2) Assume that $a \mid b$, $b \neq 0$. So,

$$b = ar, \quad r \in \mathbb{Z}.$$

Consider the case that $a \leq |b|$ is false, so $|b| < a$, or

$$-a < b < a.$$

Notice that $a = 0$ would imply that $b = 0$, but $b \neq 0$. So, $a \neq 0$. Then,

$$-1 < \frac{b}{a} = r < 1.$$

But, this would contradict the fact that $r \in \mathbb{Z}$. So, $a \leq |b|$.

(3). Suppose that $a \mid b$ and $a \mid c$, so

$$b = ar, \ c = ak, \quad r, k \in \mathbb{Z}.$$

Let $u, v \in \mathbb{Z}$,

$$bu = aru, \ cv = akv.$$

Then,

$$bu + cv = aru + akv = a(ru + kv).$$

Since $ru + kv \in \mathbb{Z}$, $a \mid bu + cv$ for all $u, v \in \mathbb{Z}$. ■.

- **Greatest common divisor:** Let a and b be integers, not both zero. Then, the **greatest common divisor (gcd)** of a and b , written (a, b) , is the largest positive integer g such that $g \mid a$ and $g \mid b$.

Notice that (a, b) is always at least 1, and $(a, b) \leq |a|$. So, only numbers from 1 to $|a|$ are considered.

- **Lemma 2.2:** Take any integers a, b with $a \neq 0$. Then,
 1. $(a, b) = (-a, b)$
 2. $(a, 0) = |a|$

Proof. Assume a, b are integers with $a \neq 0$.

(1). Assume that $(a, b) = g$